

UAV Wall — setup

A wall of live video feeds, composed onto one canvas and sent into a Pexip VMR as a single participant. This guide covers opening it, and installing the optional extras if you want recording, your own video files, or audio.

Before you start

Mac	Apple silicon (M1 or later), macOS 14 or newer. Intel Macs are not supported.
To install	Nothing. The app is self-contained. Extras are optional and covered on page 3.
For a VMR	The conference alias and, if it has one, its PIN.

Opening it the first time

The app is not signed by an identified developer, so macOS blocks it once. This is expected and only happens on the first run.

- 1 Drag **UAV Wall** to your **Applications** folder.
- 2 Double-click it. macOS refuses and says it cannot verify the developer. Click **Done**.
- 3 Open **System Settings** > **Privacy & Security**, scroll to the bottom, and click **Open Anyway** next to the UAV Wall message.
- 4 Double-click the app again and click **Open**. It will start normally every time after this.

It opens onto a working wall. Four demo clips are built in and loop continuously, so you can try the layouts and send to a VMR without any cameras or setup.

If it says the app is damaged — or nothing happens at all — macOS has quarantined it because it arrived over the network. Open **Terminal** (Applications > Utilities) and paste this one line, then open the app again:

```
xattr -dr com.apple.quarantine "/Applications/UAV Wall.app"
```

That clears the download marker. It does not change the app, and you only ever need it once.

The basics

Putting feeds on the canvas

Sources are listed down the left. Switch one on with its toggle, then use a layout button — **1-UP**, **2×2**, **3×3** or **PIP** — to place whatever is connected. Drag a tile to move it, drag its corner to resize. Double-click any tile or card to punch that feed full screen, and again to go back. **A / B / C** store layouts: click to recall, press and hold to save.

Sending it to a VMR

Type `alias@server` into **DESTINATION**, add the PIN if the VMR has one, and press **SEND TO VMR**. Everyone in the conference sees the composed canvas as one participant's video. **Settings > Canvas & sending** switches it to the presentation stream instead, so it appears alongside the participants rather than as one of them.

Adding your own sources

An IP camera (RTSP)

Settings > Feeds > **+ ADD FEED**, then type the camera's address, e.g. `rtsp://10.0.1.50:554/stream1`. Works straight away — nothing to install.

A phone, wearable or drone (RTMP/SRT)

These push video out rather than serving it, so the app runs a receiver for them. See page 4 — also nothing to install.

A video file

Settings > Feeds > **+ ADD FILE**. Needs **ffmpeg** — see page 3.

Recording

Capture any feed, or the whole canvas, to an MP4 in `~/Movies/UAV Wall`. Needs **ffmpeg** — see page 3.

Cameras with a password. Put it in the address as `rtsp://user:pass@host/path`. It is stored in plain text in the app's settings file, so do not use a camera whose password matters.

Where things are kept

Settings	<code>~/Library/Application Support/UAV Wall/uavwall.conf</code>
Recordings	<code>~/Movies/UAV Wall</code>

Installing ffmpeg

Only needed for four things: adding your own video files, recording, listening to a feed's audio, and sending a feed's audio into the VMR. Everything else — the demo clips, RTSP cameras, layouts, sending the canvas — works without it.

Why isn't it included? ffmpeg is licensed under the GPL. Bundling it would add licensing obligations to this app, so you install it yourself instead, direct from its own distributor. The app finds it automatically afterwards.

Step 1 — Install Homebrew

Homebrew installs command-line software on a Mac. Skip this if you already have it (type `brew --version` in Terminal to check).

Open **Terminal** (Applications > Utilities) and paste in one line:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

It asks for your Mac password and takes a few minutes. At the end it prints two `echo` commands under **Next steps** — run those as well, or the `brew` command will not be found.

Step 2 — Install ffmpeg

```
brew install ffmpeg
```

Several minutes, and a few hundred megabytes.

Step 3 — Restart UAV Wall

Quit the app and open it again. It looks for ffmpeg at startup, so a running app will not notice a new installation. **+ ADD FILE** and the record buttons become available.

How to tell it worked. Without ffmpeg the app shows `ffmpeg not found - import, recording and audio are off` along the bottom when it starts. With it installed, that message is gone.

Imported files are converted first

When you add a video file the app re-encodes it before use, which takes a few seconds per clip. This is deliberate: the video engine decodes some high-quality H.264 files badly, and they appear heavily blocky. The converted copies are cached, so a file added a second time is instant.

The RTMP / SRT receiver

An IP camera serves video and the wall fetches it. A phone or wearable does the opposite — it pushes video to an address you give it. The app includes a receiver for exactly that. **Nothing to install.**

- 1 Open **Settings > Feeds** and switch on **Receiver (RTMP/SRT)**. It should say **LISTENING** within a second or two.
- 2 Note the **NEXT · RTMP** address shown, e.g. `rtmp://10.0.1.12:1935/live/source-01`. Click it to copy it.
- 3 Press **+ ADD RECEIVER FEED**. A source appears in the list, ready and waiting.
- 4 In your streaming app on the phone, set that address as the destination and start streaming.
- 5 Switch the new source on. It shows `waiting for the device...` until the stream arrives, then goes live on its own.

Repeat for as many devices as you like — each press of **+ ADD RECEIVER FEED** gives out the next address. Every source's own push address is shown in the inspector at the bottom of the list when you select it; hover any line to see it in full, click to copy.

Two settings on the device that matter

Baseline profile	Set the encoder to Baseline . Higher profiles decode badly and the picture breaks up. Most apps expose this directly.
Same network	The phone must be on the same Wi-Fi as the Mac. A guest network or mobile data will not reach it.

If the device will not connect

- › Check the address on the device **exactly** — including `/live/`. Some apps reject an address without two parts after the host.
- › Give it a moment. A phone can take several seconds to negotiate; the source picks it up on its own.
- › Confirm the receiver says **LISTENING**. If it reports an error, something else on the Mac is already using the same ports.

Something to be aware of. While the receiver is on, your Mac accepts incoming video from anything on the network that knows the address. Switch it off when you have finished demonstrating.